ModelIt!

Teacher's Guide

Your Phone's Dirty Secret

The Hidden Environmental Cost of the Device in Your Pocket

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-ESS3-3, MS-ESS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-3, MS-ESS3-4

Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to trace the impact of resource extraction and consumption on Earth's systems.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — Human activities have significantly altered the biosphere, sometimes damaging or destroying natural habitats.
Crosscutting Concept	**Cause and Effect** — Students identify causal chains from consumer demand through mining to environmental damage and e-waste accumulation.

CAST Testing Alignment

MULTIPLE CHOICE:

A company mines coltan for smartphone production. Which is the most likely environmental impact of increased mining operations?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain how increasing global smartphone demand creates a chain of environmental effects from mining to e-waste.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The average smartphone contains over 30 different elements, many classified as rare earth minerals. The environmental cost of extracting, processing, manufacturing, and disposing of these devices is enormous — and growing as global demand increases.

What's Inside Your Phone

A typical smartphone contains: lithium (battery), cobalt (battery), gold and silver (circuit connections), rare earth elements like neodymium (speakers/vibration motor), indium (touchscreen), tantalum from coltan (capacitors), and silicon (processor). Many of these are mined in environmentally sensitive areas.

The Mining Problem

Rare earth mineral mining is extremely destructive. It involves stripping large areas of land, using toxic chemicals to extract minerals, generating radioactive waste, and contaminating water supplies. The Democratic Republic of Congo produces 60% of the world's cobalt, often using child labor in dangerous conditions.

The E-Waste Crisis

The world generates approximately 50 million tons of e-waste per year. Only about 20% is formally recycled. The rest ends up in landfills or is shipped to developing countries where it's dismantled by hand, exposing workers to lead, mercury, cadmium, and other toxins.

Solutions and Alternatives

Strategies include: extending device lifespan (repair instead of replace), improving recycling technology, designing devices for disassembly, using recycled materials in new devices, and developing alternative materials. The 'circular economy' model aims to eliminate waste entirely.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify consumer demand, mining intensity, environmental damage, and e-waste accumulation as system components.

E — Establish Relationships
Students determine that demand drives mining (positive), mining drives damage (positive), and demand drives e-waste (positive).

V — Visualize & Model
Students build the lifecycle model showing how consumption drives a chain of environmental effects.

E — Evaluate Outcomes
Students compare current trends, maximum consumption, and circular economy scenarios.

R — Revise & Extend
Students add recycling rate or carbon emissions to create a more complete lifecycle model.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at your phone. Do you know what it took to make it?"*

Watch For

Ask students to hold up their phones. Create cognitive dissonance.

Slide 2: Learning Objectives

3 min

Say: *"Today we trace the hidden journey of your phone — from mine to pocket to landfill."*

Watch For

Define lifecycle analysis with a simple example first.

Slide 3: Big Question

3 min

Say: *"Your phone cost a few hundred dollars. But what's the REAL cost?"*

Watch For

Quick write: List everything you think goes into making a phone.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the environmental impact chain."*

Watch For

Preview how we'll trace cause and effect.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of the phone's environmental impact system?"*

Watch For

Sort components. Discuss what students can vs. can't control.

Slide 6: Activity 2: Connections

8 min

Say: *"Every new phone purchase connects to a chain of environmental effects."*

Watch For

Map the cause-and-effect chain from demand to damage.

Slide 7: Activity 3: Simulation

10 min

Say: *"What if everyone upgraded every year? What if we all kept phones twice as long?"*

Watch For

Run all three scenarios. Calculate the difference.

Slide 8: Discoveries

5 min

Say: *"What surprised you most about your phone's environmental footprint?"*

Watch For

Discuss personal responsibility vs. systemic change.

Slide 9: STEM Challenge

12 min

Say: *"Your school district hired you to design a sustainable technology program."*

Watch For

Teams design evidence-based programs. Present to class.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Consumer Demand, Mining Intensity
- **INTERNAL:** Environmental Damage, E-Waste Accumulation

Relationships

- **Consumer Demand to Mining Intensity:** POSITIVE (+) — More demand for phones means more minerals need to be extracted from the Earth.
- **Mining Intensity to Environmental Damage:** POSITIVE (+) — More mining means more land destruction, water pollution, and toxic waste.
- **Consumer Demand to E-Waste Accumulation:** POSITIVE (+) — More phones purchased means more phones eventually discarded as e-waste.

Key Discoveries

- A single smartphone contains over 30 different minerals from mines across the world
- Mining rare earth minerals generates toxic waste that contaminates water and soil
- Only about 20% of e-waste is properly recycled — the rest goes to landfills or developing countries
- Reducing demand (keeping phones longer) has a bigger impact than recycling alone

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Recycling solves the e-waste problem"	Only 20% of e-waste is properly recycled. Even recycled electronics lose material in the process. REDUCING consumption (keeping devices longer) is far more effective than recycling alone.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"My phone doesn't affect the environment"	Every phone requires mining tons of rock, uses energy from fossil fuels to manufacture, and generates toxic waste at end of life. Multiply by 1.5 billion phones sold per year and the impact is massive.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Technology is always progress"	Technology provides amazing benefits, but it also has environmental costs that we need to account for. The goal isn't to stop using technology, but to use it more responsibly and demand sustainable practices.